

AER1516: Robot Motion Planning

Assignment #1: Some Background Theory

Student: Azure Lin

1. Euler Angles

Consider the Z–Y–X Euler angle sequence used to represent an element of $SO(3)$ (as shown on Slide 14 of Lecture 2 and in the accompanying note). What happens when $Y = +\frac{\pi}{2}$ radians or $Y = -\frac{\pi}{2}$ radians, and why is this a concern?

Answer. At $Y = \beta = \pm\frac{\pi}{2}$, the Euler angle parameterization becomes singular: the first and third rotation axes become colinear, so one rotational degree of freedom is lost (gimbal lock, Fig. 1), and the mapping between Euler rates and angular velocity becomes non-invertible, causing numerical ill-conditioning in estimation and control.

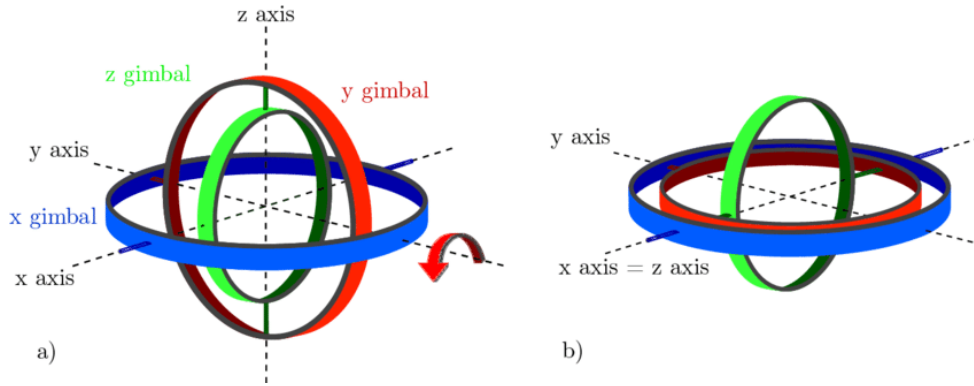


Figure 1: Illustration of gimbal lock from Zeitlhöfler Zeitlhöfler (2019).

What happens when $Y = \beta = \pm\frac{\pi}{2}$. Let $F_{\rightarrow 1}$ be the world frame and $F_{\rightarrow 2}$ the body frame. Let $r_1, r_2 \in \mathbb{R}^3$ be the coordinates of the same geometric vector $\vec{r}$ expressed in $F_{\rightarrow 1}$ and $F_{\rightarrow 2}$. Define C_{12} by $r_1 = C_{12}r_2$, where

$$C_{12} = R_Z(\alpha) R_Y(\beta) R_X(\gamma)$$

$$= \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \gamma & -\sin \gamma \\ 0 & \sin \gamma & \cos \gamma \end{bmatrix}$$

The angular velocity is the sum of the three instantaneous rotations about their current axes:

$$\omega_{\rightarrow 21} = \dot{\alpha} \vec{z}_{\rightarrow 1} + \dot{\beta} \left(R_Z(\alpha) \vec{y}_{\rightarrow 1} \right) + \dot{\gamma} \left(R_Z(\alpha) R_Y(\beta) \vec{x}_{\rightarrow 1} \right),$$

$$\Rightarrow \omega_1^{21} = \dot{\alpha} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \dot{\beta} R_Z(\alpha) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \dot{\gamma} R_Z(\alpha) R_Y(\beta) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} 0 & -\sin \alpha & \cos \alpha \cos \beta \\ 0 & \cos \alpha & \sin \alpha \cos \beta \\ 1 & 0 & -\sin \beta \end{bmatrix}}_{\text{Jacobian matrix } J(\alpha, \beta)} \begin{bmatrix} \dot{\alpha} \\ \dot{\beta} \\ \dot{\gamma} \end{bmatrix}.$$

At $Y = \beta = \pm \frac{\pi}{2}$ we have $\cos \beta = 0$, so the determinant of the Jacobian matrix is 0

$$\det(J(\alpha, \beta)) \Big|_{\beta=\pm \frac{\pi}{2}} = \det \begin{bmatrix} 0 & -\sin \alpha & 0 \\ 0 & \cos \alpha & 0 \\ 1 & 0 & -\sin \beta \end{bmatrix} = 0.$$

Therefore $J(\alpha, \beta)$ is not invertible, and the Euler rates $(\dot{\alpha}, \dot{\beta}, \dot{\gamma})$ cannot be uniquely recovered from a given angular velocity $\omega_{\rightarrow 21}$.

Geometrically, the current x -axis in body frame before applying $R_X(\gamma)$ is

$$R_Z(\alpha)R_Y(\beta) x_{\rightarrow 1}$$

which becomes parallel to the z -axis in world frame:

$$R_Z(\alpha)R_Y\left(\frac{\pi}{2}\right) x_{\rightarrow 1} = -z_{\rightarrow 1}, \quad R_Z(\alpha)R_Y\left(-\frac{\pi}{2}\right) x_{\rightarrow 1} = +z_{\rightarrow 1}.$$

Thus yaw α and roll γ generate the same physical rotation direction, which causes a loss of one degree of freedom, this is known as gimbal lock.

Why this is a problem. Near $\beta = \pm \frac{\pi}{2}$ the mapping becomes ill-conditioned, so small attitude changes can cause large swings in (α, γ) and unstable numerical behaviour in estimation and control.

2. Dense Warehouse Graph

A delivery robot operates in a large warehouse represented as a weighted graph $G = (V, E)$, where V is the set of n intersections and E is the set of m hallways connecting them. Edge weights represent traversal distances. Also, the warehouse layout is highly connected, so the graph is dense, with $m = \Theta(n^2)$.

(a) Space complexity. The robot has limited onboard memory and must store a representation of the map. Compare the space complexity of an adjacency matrix and an adjacency list¹ for this graph. Express the memory usage using Big-O notation in terms of n and m , and determine which representation is asymptotically more appropriate for a dense graph.

Answer (a).

Adjacency matrix. Store the map as an $n \times n$ table A , where $A[i, j]$ stores the edge weight w_{ij} if $(i, j) \in E$, and a sentinel value (e.g., ∞ or None) otherwise. This requires n^2 entries, hence the space complexity is $O(n^2)$.

Adjacency list. Store an array of n lists (one per vertex), where each list contains neighbor-records of the form (j, w_{ij}) . The n list headers cost $O(n)$ space.

- for a directed graph, each edge appears once across all lists, contributing m records.
- for an undirected graph, each edge appears twice, contributing $2m$ records.

Thus the total space is $O(n + 2m) = O(n + m)$.

Dense case. Given $m = \Theta(n^2)$,

$$O(n + m) = O\left(n + \Theta(n^2)\right) = O(n^2).$$

Therefore, for dense graphs, adjacency matrices and adjacency lists are asymptotically tied in space usage $\Theta(n^2)$. In practice, an adjacency matrix is preferred in dense graphs due to its $O(1)$ edge lookup time.

¹you will need to look up the definition of an adjacency list.

(b) Greedy path strategy. Consider the following greedy strategy for finding a path from the Start vertex to the Goal vertex:

At each step, move to the adjacent vertex connected by the smallest-weight outgoing edge.

Is this strategy guaranteed to find the shortest path in a weighted graph? If it is not guaranteed to do so, construct a counterexample and explain why the strategy fails.

Answer (b). No, greedy algorithm is not guaranteed to find a shortest path because it optimizes only the next edge (local cost) rather than the total path cost (global sum).

Counterexample. Let the vertices be S (Start), A , B , and G (Goal), with directed edges:

$$w(S, A) = 1, \quad w(S, B) = 2, \quad w(A, G) = 100, \quad w(B, G) = 2.$$

The greedy rule picks the smallest outgoing edge from the current vertex. From S , it chooses $S \rightarrow A$ (since $1 < 2$), then it must take $A \rightarrow G$. So the greedy path has cost $1 + 100 = 101$. However, the path $S \rightarrow B \rightarrow G$ has cost $2 + 2 = 4$, which is shorter. This fails because the greedy rule is local: it minimizes the next edge weight, not the total remaining cost to reach G .

(c) Permutation algorithm. An intern proposes an alternative algorithm that enumerates all possible permutations of intersections to find the shortest path, resulting in a time complexity of $O(n!)$. Is this a polynomial-time algorithm? Justify your answer by comparing the growth rate of $n!$ to n^k for an arbitrary fixed constant k .

Answer (c). No. A running time is polynomial-time if and only if $T(n) = O(n^k)$ for some fixed constant k . We show $n! \notin O(n^k)$ for every fixed k by proving

$$\lim_{n \rightarrow \infty} \frac{n!}{n^k} = \infty.$$

For $n \geq 2$,

$$n! = \prod_{i=1}^n i \geq \prod_{i=\lceil n/2 \rceil}^n i.$$

There are $n - \lceil n/2 \rceil + 1 = \lfloor n/2 \rfloor + 1$ terms in this product, and each term is at least $n/2$, so

$$\prod_{i=\lceil n/2 \rceil}^n i \geq \left(\frac{n}{2}\right)^{\lfloor n/2 \rfloor + 1} \geq \left(\frac{n}{2}\right)^{n/2}.$$

Hence

$$\frac{n!}{n^k} \geq \frac{(n/2)^{n/2}}{n^k} = \frac{n^{n/2-k}}{2^{n/2}}.$$

Taking logs,

$$\ln\left(\frac{n^{n/2-k}}{2^{n/2}}\right) = \left(\frac{n}{2} - k\right) \ln n - \frac{n}{2} \ln 2 \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

so

$$\lim_{n \rightarrow \infty} \frac{n!}{n^k} \rightarrow \infty$$

Therefore $O(n!)$ is not polynomial time.

3. Unit Circle Charts

The unit circle S^1 is defined by the set of points

$$S^1 \triangleq \{(x, y) \mid x^2 + y^2 = 1\},$$

and cannot be covered by a single chart (mapping to an open set in $\mathbb{R}$). However, it can be covered by two charts. Define two charts, (U, φ) and (V, ψ) , such that $U \subset S^1$, $\varphi : U \rightarrow \mathbb{R}$ and $V \subset S^1$, $\psi : V \rightarrow \mathbb{R}$.

Show that the two charts cover S^1 and that the map $\varphi \circ \psi^{-1}$ is a diffeomorphism. Be sure to carefully indicate the points that are in the sets U and in V .

Answer.

Idea. We build coordinates on most of the circle by stereographic projection. Fix a pole (north or south). For any point P on the circle excluding the pole, draw the straight line from the pole through P , where that line hits the x -axis is a real number. That real number is the coordinate of P in that chart.

Let the north pole be $N = (0, 1)$ and the south pole be $S = (0, -1)$. Define

$$U := S^1 \setminus \{N\}, \quad V := S^1 \setminus \{S\}.$$

Then U and V are open in S^1 and

$$U \cup V = (S^1 \setminus \{N\}) \cup (S^1 \setminus \{S\}) = S^1 \setminus (\{N\} \cap \{S\}) = S^1,$$

so **the two chart domains cover S^1** .

Chart from the north pole. Take $P = (x, y) \in U = S^1 \setminus \{N\}$ with $N = (0, 1)$.

1. Parametrize the line through N and P :

$$L_N(\lambda) = N + \lambda(P - N) = (0, 1) + \lambda(x, y - 1) = (\lambda x, 1 + \lambda(y - 1)).$$

2. Intersect with the x -axis: set the y -coordinate to 0:

$$1 + \lambda(y - 1) = 0 \implies \lambda = \frac{1}{1 - y}.$$

This is well-defined because $P \neq N \implies y \neq 1 \implies 1 - y \neq 0$.

3. Read off the x -coordinate of the intersection:

$$\varphi(x, y) := \lambda x = \frac{x}{1 - y}.$$

Inverse map φ^{-1} . Given $t \in \mathbb{R}$, we solve for $(x, y) \in S^1 \setminus \{N\}$ such that

$$t = \varphi(x, y) = \frac{x}{1 - y}, \quad x^2 + y^2 = 1.$$

From $t = \frac{x}{1 - y}$ we have $x = t(1 - y)$. Substitute into $x^2 + y^2 = 1$:

$$\begin{aligned} t^2(1 - y)^2 + y^2 &= 1 \\ t^2(1 - 2y + y^2) + y^2 &= 1 \\ (t^2 + 1)y^2 - 2t^2y + (t^2 - 1) &= 0. \end{aligned}$$

Solve this quadratic in y . The discriminant is

$$\Delta = (-2t^2)^2 - 4(t^2 + 1)(t^2 - 1) = 4t^4 - 4(t^4 - 1) = 4,$$

so

$$y = \frac{2t^2 \pm 2}{2(t^2 + 1)} = \frac{t^2 \pm 1}{t^2 + 1}.$$

The $+$ root gives $y = 1$, i.e. the excluded north pole N , so we take

$$y = \frac{t^2 - 1}{t^2 + 1}.$$

Then

$$x = t(1 - y) = t \left(1 - \frac{t^2 - 1}{t^2 + 1} \right) = t \left(\frac{2}{t^2 + 1} \right) = \frac{2t}{t^2 + 1}.$$

Hence the inverse map is

$$\varphi^{-1}(t) = \left(\frac{2t}{t^2 + 1}, \frac{t^2 - 1}{t^2 + 1} \right).$$

and $\varphi : U \rightarrow \mathbb{R}$ is a diffeomorphism.

Chart from the south pole. Take $P = (x, y) \in V = S^1 \setminus \{S\}$ with $S = (0, -1)$.

1. Parametrize the line through S and P :

$$L_S(\mu) = S + \mu(P - S) = (0, -1) + \mu(x, y + 1) = (\mu x, -1 + \mu(y + 1)).$$

2. Intersect with the x -axis:

$$-1 + \mu(y + 1) = 0 \implies \mu = \frac{1}{1 + y}.$$

This is well-defined because $P \neq S \implies y \neq -1 \implies 1 + y \neq 0$.

3. Define the coordinate map:

$$\psi(x, y) := \mu x = \frac{x}{1 + y}.$$

Similarly, ψ is a diffeomorphism

$$\psi^{-1}(t) = \left(\frac{2t}{t^2 + 1}, \frac{1 - t^2}{t^2 + 1} \right).$$

Transition map on the overlap. The overlap is

$$U \cap V = S^1 \setminus \{N, S\}.$$

On $U \cap V$ we have $x \neq 0$ (since $x = 0$ on S^1 implies $(x, y) \in \{N, S\}$), hence

$$\psi(U \cap V) = \varphi(U \cap V) = \mathbb{R} \setminus \{0\}.$$

For $(x, y) \in U \cap V$, using $x^2 + y^2 = 1$,

$$\varphi(x, y) \psi(x, y) = \frac{x}{1 - y} \cdot \frac{x}{1 + y} = \frac{x^2}{1 - y^2} = \frac{x^2}{x^2} = 1.$$

Therefore on $U \cap V$,

$$\varphi(x, y) = \frac{1}{\psi(x, y)}$$

so for $t \in \mathbb{R} \setminus \{0\}$,

$$(\varphi \circ \psi^{-1})(t) = \frac{1}{t}.$$

Since $t \mapsto 1/t$ is smooth on $\mathbb{R} \setminus \{0\}$ and its inverse is itself (also smooth), the map $\varphi \circ \psi^{-1} : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$ is a diffeomorphism.

4. Unit Sphere Metric

Consider the unit sphere S^2 embedded in $\mathbb{R}^3$. What is an appropriate distance metric on the unit sphere? That is, how would you define the line element on the surface? (Hint: this question is easier than it looks—consider the coordinates you might use to represent points on the sphere and go from there...)

Answer.

Define line element. Use spherical coordinates (θ, ϕ) on the unit sphere:

$$\mathbf{x}(\theta, \phi) = \begin{bmatrix} x(\theta, \phi) \\ y(\theta, \phi) \\ z(\theta, \phi) \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{bmatrix}, \quad \theta \in (0, \pi), \phi \in (0, 2\pi).$$

Since S^2 is embedded in $\mathbb{R}^3$ with Euclidean inner product, the squared infinitesimal arc length along the surface is the dot product in $\mathbb{R}^3$:

$$ds^2 := d\mathbf{x} \cdot d\mathbf{x}$$

Substitute with the surface differential $d\mathbf{x} = \mathbf{x}_\theta d\theta + \mathbf{x}_\phi d\phi$:

$$\begin{aligned} ds^2 &= (\mathbf{x}_\theta d\theta + \mathbf{x}_\phi d\phi) \cdot (\mathbf{x}_\theta d\theta + \mathbf{x}_\phi d\phi) \\ &= (\mathbf{x}_\theta \cdot \mathbf{x}_\theta) d\theta^2 + 2(\mathbf{x}_\theta \cdot \mathbf{x}_\phi) d\theta d\phi + (\mathbf{x}_\phi \cdot \mathbf{x}_\phi) d\phi^2. \end{aligned}$$

Thus we need the three dot products $\mathbf{x}_\theta \cdot \mathbf{x}_\theta$, $\mathbf{x}_\theta \cdot \mathbf{x}_\phi$, and $\mathbf{x}_\phi \cdot \mathbf{x}_\phi$.

$$\begin{aligned} \mathbf{x}_\theta(\theta, \phi) &:= \frac{\partial \mathbf{x}}{\partial \theta} = \begin{bmatrix} \frac{\partial}{\partial \theta}(\sin \theta \cos \phi) \\ \frac{\partial}{\partial \theta}(\sin \theta \sin \phi) \\ \frac{\partial}{\partial \theta}(\cos \theta) \end{bmatrix} = \begin{bmatrix} \cos \theta \cos \phi \\ \cos \theta \sin \phi \\ -\sin \theta \end{bmatrix}, \\ \mathbf{x}_\phi(\theta, \phi) &:= \frac{\partial \mathbf{x}}{\partial \phi} = \begin{bmatrix} \frac{\partial}{\partial \phi}(\sin \theta \cos \phi) \\ \frac{\partial}{\partial \phi}(\sin \theta \sin \phi) \\ \frac{\partial}{\partial \phi}(\cos \theta) \end{bmatrix} = \begin{bmatrix} -\sin \theta \sin \phi \\ \sin \theta \cos \phi \\ 0 \end{bmatrix}. \end{aligned}$$

(i) Compute $\mathbf{x}_\theta \cdot \mathbf{x}_\theta$.

$$\begin{aligned} \mathbf{x}_\theta \cdot \mathbf{x}_\theta &= (\cos \theta \cos \phi)^2 + (\cos \theta \sin \phi)^2 + (-\sin \theta)^2 \\ &= \cos^2 \theta \cos^2 \phi + \cos^2 \theta \sin^2 \phi + \sin^2 \theta \\ &= \cos^2 \theta (\cos^2 \phi + \sin^2 \phi) + \sin^2 \theta \\ &= \cos^2 \theta (1) + \sin^2 \theta \\ &= 1. \end{aligned}$$

(ii) Compute $\mathbf{x}_\theta \cdot \mathbf{x}_\phi$.

$$\begin{aligned} \mathbf{x}_\theta \cdot \mathbf{x}_\phi &= (\cos \theta \cos \phi)(-\sin \theta \sin \phi) + (\cos \theta \sin \phi)(\sin \theta \cos \phi) + (-\sin \theta)(0) \\ &= -\cos \theta \sin \theta \cos \phi \sin \phi + \cos \theta \sin \theta \sin \phi \cos \phi \\ &= 0. \end{aligned}$$

(iii) Compute $\mathbf{x}_\phi \cdot \mathbf{x}_\phi$.

$$\begin{aligned} \mathbf{x}_\phi \cdot \mathbf{x}_\phi &= (-\sin \theta \sin \phi)^2 + (\sin \theta \cos \phi)^2 + 0^2 \\ &= \sin^2 \theta \sin^2 \phi + \sin^2 \theta \cos^2 \phi \\ &= \sin^2 \theta (\sin^2 \phi + \cos^2 \phi) \\ &= \sin^2 \theta (1) \\ &= \sin^2 \theta. \end{aligned}$$

Using the computed dot products:

$$(\mathbf{x}_\theta \cdot \mathbf{x}_\theta) = 1, \quad (\mathbf{x}_\theta \cdot \mathbf{x}_\phi) = 0, \quad (\mathbf{x}_\phi \cdot \mathbf{x}_\phi) = \sin^2 \theta,$$

the expansion becomes

$$\begin{aligned} ds^2 &= 1 \cdot d\theta^2 + 2 \cdot 0 \cdot d\theta d\phi + \sin^2 \theta d\phi^2 \\ &= d\theta^2 + \sin^2 \theta d\phi^2. \end{aligned}$$

Conclusion. An appropriate distance metric (line element) on the unit sphere is

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2$$

Equivalently, the infinitesimal arc length is

$$ds = \sqrt{d\theta^2 + \sin^2 \theta d\phi^2}.$$

5. Wavefront Planner

Does the wavefront planner on a discrete grid yield the shortest distance to the goal? Why or why not? If so, what metric is the wavefront planner using?

Answer. It depends on the edge-cost structure.

Represent the discretised free space as a graph $G = (V, E)$, where V is the set of free cells and E connects adjacent free cells. Let the traversal cost be a function $c : E \rightarrow \mathbb{R}_{\geq 0}$. Define the (discrete) distance-to-goal as the shortest-path cost

$$D(v) = \min_{P: v \rightarrow g} \sum_{e \in P} c(e),$$

where P ranges over all paths from cell v to the goal cell g .

- **Uniform costs (unweighted grid).** If $c(e) \equiv 1$ for all $e \in E$, then for any path P , the total cost equals the number of moves. Hence

$$D(v) = \min_{P: v \rightarrow g} |P|.$$

In this case, the wavefront planner works like breadth-first search (BFS) started from the goal: it expands cells in FIFO “layers” and labels each reachable cell with the minimum number of moves to g . Therefore, the wavefront planner does yield the shortest distance-to-goal under this (unit-cost) grid metric.

Metric (uniform-cost, obstacle-free grid).

- 4-neighbour connectivity: D reduces to the **Manhattan** distance L_1 .
- 8-neighbour connectivity with unit-cost diagonals: D reduces to the **Chebyshev** distance L_∞ .

With obstacles, D remains the shortest-path metric on the free-cell graph induced by the same adjacency.

- **Non-uniform costs (weighted grid).** If edge costs vary (e.g. diagonal moves cost $\sqrt{2}$), a wavefront planner is **not** guaranteed to minimise $\sum_{e \in P} c(e)$, because it expands nodes by hop-count rather than by nondecreasing accumulated cost, and can therefore finalise a cell before a cheaper route to it is discovered. To obtain an optimal distance field under weighted costs, the wavefront expansion must be performed in order of increasing cost-to-come, i.e. **Dijkstra’s algorithm**.

References

Julian Zeitlhöfler. Nominal and observation-based attitude realization for precise orbit determination of the jason satellites. Master’s thesis, Technical University of Munich, Munich, Germany, June 2019. URL <https://mediatum.ub.tum.de/doc/1535899/1535899.pdf>.